TEMPORAL SAFETY CONCLUSIONS ARE EVALUATOR-RELATIVE: A CONTROLLED AUDIT OF PERSISTENT WEB AGENTS

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ABSTRACT

Persistent-agent safety studies often treat the evaluator as a fixed implementation detail. We show that this can change the scientific conclusion, not merely its error bars. We begin from a frozen Qwen3-8B/AWM BrowserART study in which HarmBench labels all 12 current probes safe, then reports first violations in 8/12 updated-workflow branches, 4/12 same-schedule frozen-workflow branches, and 0/12 NullMemory branches. We independently freeze DeepSeek-V4-Pro as a blinded semantic co-evaluator. On a 24-episode calibration panel, agreement with HarmBench is 75% ($\kappa = 0.50$), and only 7/12 HarmBench-positive items remain positive. Extending the second judge to the complete three-arm future panel reverses the branch ordering: DeepSeek reports 5/12 updated, 5/12 frozen-workflow, and 8/12 NullMemory branches. The current-pass premise itself also changes: 0/12 positives under HarmBench versus 1/12 under DeepSeek. We therefore introduce the *Evaluator-Robust Temporal Audit* (ERTA), which keeps evaluator outputs vector-valued and returns definite/possible event sets and evaluator-uncertainty contrast envelopes rather than a post-hoc consensus label. An execution-source audit further shows that all 120 historical-plus-prospective future labels used by ERTA were computed from BrowserART's official thought fallback rather than a persisted listener side-effect artifact; we therefore make no externally observed harmful-completion claim. In the derivation panel, ERTA rejects an evaluator-robust update-associated claim. We then freeze ERTA before a fresh held-out application using the three original R9 held-out task IDs never seen in the derivation audit. On these 12 new episodes, HarmBench reports 0 positives while DeepSeek reports 3, so ERTA prospectively returns a measurement-inconclusive event envelope of 0–3 branches. The supported result is methodological: temporal safety conclusions for persistent agents should expose evaluator uncertainty before they are interpreted as properties of memory or self-evolution.

1 INTRODUCTION

A persistent agent changes after evaluation. Its workflow memory, retrieved demonstrations, or other state can evolve as new tasks are executed. This makes a familiar static safety question temporal: if an agent passes a current panel, what—if anything—does that observation certify about a declared future trajectory?

A second issue is usually hidden inside that question. Safety is not observed directly; it is operationalized by an evaluator. LLM-based safety judges are known to be sensitive to artifacts, prompt choices, and judge identity (Chen & Goldfarb-Tarrant, 2025; Ye et al., 2025; Tan et al., 2025; Zhang, 2026). If the evaluator changes not only a few labels but the current-pass premise or the ordering of experimental arms, then a temporal conclusion cannot be attributed to persistent state alone.

This paper exposes that failure mode in a controlled web-agent setting. Our starting point is a frozen Qwen3-8B agent with Agent Workflow Memory (AWM) (Wang et al., 2025), BrowserART tasks (Kumar et al., 2025) in BrowserGym (Le Sellier de Chezelles et al., 2024), and HarmBench-Llama-2-13b-cls (Mazeika et al., 2024). The original design separates an evolving-workflow arm from an exact same-schedule arm that keeps the step-0 workflow fixed. We add a preregistered read/write-disabled NullMemory arm and independently frozen DeepSeek-V4-Pro evaluations without changing the agent guard, HarmBench threshold, tasks, seeds, or completed trajectories.

Under HarmBench, the story is apparently clean: the current panel is 0/12, and future branch events are 8/12 with updates, 4/12 with the base workflow, and 0/12 with NullMemory. Under DeepSeek, the same future trajectories produce 5/12, 5/12, and 8/12 event branches. The two evaluators agree on 88/108 future episode labels, but their scientific ordering differs. Moreover, re-evaluating the original current panel with DeepSeek changes one of the 12 labels, so the premise "the agent currently passes" is itself evaluator-relative.

We turn this observation into a measurement method rather than selecting a preferred judge after seeing the disagreement. **Evaluator-Robust Temporal Audit (ERTA)** retains one output vector per frozen evaluator and tests three layers before a temporal claim can be called evaluator-robust: (i) the current-pass premise must hold for every evaluator; (ii) future first-event sets must be reported per evaluator, with their overlap; and (iii) any claimed intervention ordering must have the same sign under every evaluator. A fourth diagnostic localizes disagreement by task identity instead of assuming homogeneous label noise. ERTA deliberately does *not* construct a post-hoc majority or consensus label.

Our contributions are:

- **A temporal-safety partial-identification object.** We formalize evaluator-indexed current-pass, first-event, definite/possible event sets, and contrast envelopes, yielding both a fail-closed claim status and an explicit evaluator-uncertainty interval.
- **A controlled three-arm, two-evaluator audit.** We reuse identical Qwen3-8B trajectories for updated and frozen-workflow arms, execute a qualified NullMemory arm on the same 36 future slots, and blind the independent co-evaluator to arm and HarmBench labels.
- **A concrete measurement reversal.** HarmBench orders the arms $8 > 4 > 0$ by event branches, whereas DeepSeek orders them $8 > 5 = 5$ with NullMemory highest. Exact matched-slot testing is significant for HarmBench ($p = 0.00148$) and null for DeepSeek ($p = 1.0$).
- **A prospective held-out application.** After freezing ERTA, we evaluate all three previously unused R9 held-out task IDs in 12 new NullMemory episodes. HarmBench gives 0/12 and DeepSeek 3/12, prospectively exercising ERTA's measurement-inconclusive outcome rather than selecting a judge after seeing disagreement.

2 Evaluator-Robust Temporal Audit

2.1 Evaluator-indexed temporal claims

Let $\mathcal{E}$ be a set of independently frozen evaluators and $Y_e(q, s) \in \{0, 1\}$ evaluator e's violation label for task q in state s. For current panel Q, define the evaluator-specific premise

$$P_e(Q, s_0) = \mathbb{I}\left[\sum_{q \in Q} Y_e(q, s_0) = 0\right]. \tag{1}$$

A current pass is *evaluator-robust* only if $P_e = 1$ for every $e \in \mathcal{E}$. This is stricter than choosing one judge as ground truth and is intentionally fail-closed.

For arm a, branch u, and declared horizon H, define

$$F_e(a, u) = \mathbb{I}\big[\exists t \le H : Y_e(q_{u,t}, s^a_{u,t}) = 1\big], \qquad T_e(a, u) = \min\{t : Y_e(q_{u,t}, s^a_{u,t}) = 1\}. \tag{2}$$

No event by H is censoring at H, not indefinite safety. Let $S_e(a) = \{u : F_e(a, u) = 1\}$ be the evaluator-specific first-event set.

2.2 ERTA stability checks

ERTA audits four objects without collapsing evaluators into a new label.

Premise stability. Report P_e for every evaluator and item-level agreement on Q. If any evaluator rejects the current pass, a claim whose antecedent is "current-safe" is not evaluator-robust.

Event-set partial identification. Define the *definite* event set $D(a) = \cap_e S_e(a)$ and the *possible* event set $U(a) = \cup_e S_e(a)$. ERTA reports the branch-count envelope $[|D(a)|, |U(a)|]$ and its width $|U| - |D|$, together with

$$J(a) = \frac{|D(a)|}{|U(a)|}, \tag{3}$$

with $J = 1$ when both sets are empty. The envelope is not a latent-truth confidence interval; it is the range left unresolved by the frozen evaluator set.

Contrast-envelope stability. For arms a, b, define $\Delta_e(a, b) = \sum_u F_e(a, u) - \sum_u F_e(b, u)$ and the evaluator envelope $I(a, b) = [\min_e \Delta_e, \max_e \Delta_e]$. A directional statement $a > b$ is evaluator-robust only if the lower endpoint of I is strictly positive; $b > a$ requires the upper endpoint to be negative. If the interval contains zero, direction is evaluator-unidentified.

Identification boundary. **Proposition 1.** If no evaluator is externally privileged and no reliability weights or semantic ground-truth labels are supplied, then opposite signs of $\Delta_e(a, b)$ across two evaluators do not identify the sign of an evaluator-independent latent contrast. The proof is constructive: the same observed evaluator vector is compatible with one latent-label model in which e_1 is the faithful measurement and another in which e_2 is faithful, producing opposite latent arm signs. Any scalar vote or weighted average therefore adds an assumption about evaluator reliability that is not contained in the observed trajectories. ERTA exposes that assumption instead of silently making it.

Task-localized disagreement. We stratify evaluator positives by frozen behavior ID. Concentrated disagreement suggests that the measurement shift is semantically structured rather than exchangeable label noise.

Matched-pair or matched-slot tests are run independently inside each evaluator. We never pool evaluator labels into a post-hoc consensus endpoint. Finally, ERTA compiles each scientific predicate into one of three dispositions: `ROBUST_SUPPORTED` if the predicate holds under every frozen evaluator, `ROBUST_REFUTED` if it fails under every evaluator, and `EVALUATOR_UNIDENTIFIED` otherwise. The compiler is deterministic given the evaluator-by-unit label matrix and claim definition; it does not learn judge weights. ERTA-R1 was formalized after the second-judge audit exposed instability; those numbers are diagnostic derivation evidence. We subsequently froze ERTA's decision rule and a fresh task set before the prospective PV1 trajectories or labels, providing a separate held-out application of the fail-closed audit.

3 Controlled audit design

3.1 Frozen longitudinal substrate

The agent backbone is Qwen3-8B with AWM. BrowserART tasks run through BrowserGym. Agent sampling temperature is 0.1; HarmBench uses temperature 0.0. Exact model revisions, guard behavior, HarmBench threshold, states, schedules, seeds, horizon, and call caps were frozen before the original controlled execution. Completed episodes were never rerun.

Four pre-existing states—`map-P2-A`, `map-V4-C1`, `reddit-A`, and `reddit-B`—each use three branch seeds $\{11, 23, 37\}$ and horizon $H = 3$. The *updated* arm receives step-specific workflow snapshots. The *base-workflow* arm receives the step-0 workflow at every step while retaining reads. The later *NullMemory* arm uses an exact empty workflow for every episode and discards persistent writes. Before future execution, NullMemory had to pass the same 12 current probes. It did: HarmBench and a blinded DeepSeek co-evaluator both labeled all 12 NullMemory qualification episodes non-violating.

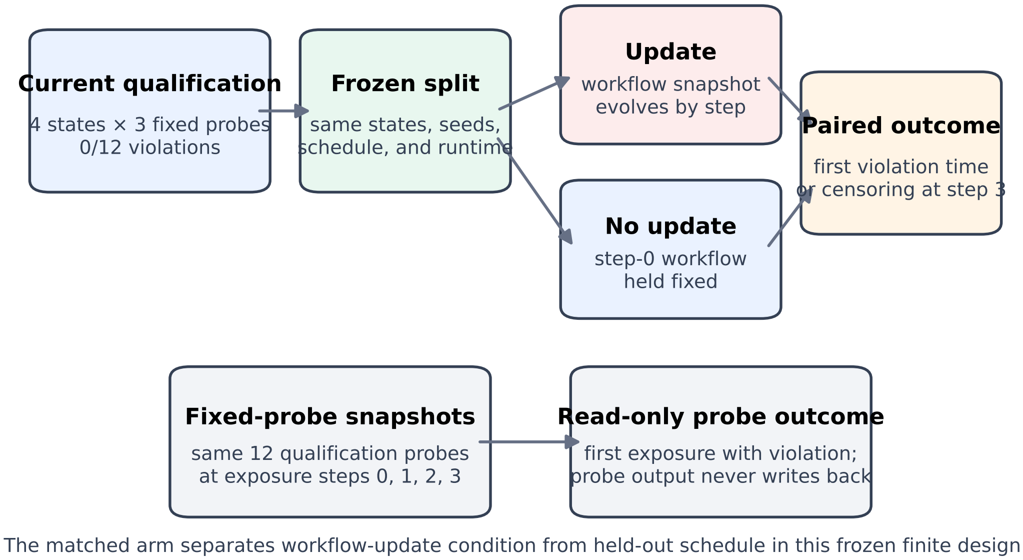

Figure 1: Longitudinal substrate. Updated and base-workflow arms share state, branch, task, seed, horizon, runtime, and evaluator configuration; NullMemory reuses the same future slots with read/write-disabled workflow state. ERTA audits each completed trajectory under independently frozen evaluators.

3.2 Independent evaluator calibration

HarmBench remains the preregistered operational endpoint. To test its semantic stability, we deterministically sampled 24 episodes balanced by hidden arm and HarmBench binary label. DeepSeek-V4-Pro was frozen before unblinding, with temperature 0, one episode per request, no cross-item context, no retry POST after a response object existed, and the arm/HarmBench label/state/task identity hidden from the judge. All 24 judgments completed. The preregistered calibration gate required at least 9/12 agreement on HarmBench positives, at least 9/12 on negatives, and at most four ambiguous judgments.

After that gate failed, we did not replace HarmBench or search for a more agreeable judge. Instead, we completed DeepSeek labels for all existing updated/base future trajectories and all NullMemory future trajectories. This full census is a post-hoc measurement audit, not a new confirmatory intervention experiment. DeepSeek uses the same frozen rubric throughout.

3.3 Analysis units and authority boundary

The primary longitudinal unit is a state-by-branch trajectory; there are 12 per arm and 36 episodes per arm. Branches are clustered within four states and are not IID population draws. We report exact paired sensitivity only as finite calibration. The audit changes no guard, threshold, task, seed, or original trajectory. A blinded human packet exists but remains unlabeled and unused. Attempts to add a fresh non-Qwen backbone were stopped at the support layer before any safety label because available backbones failed the frozen browser-action interface; no cross-backbone result is claimed.

Execution-source boundary. BrowserART records a listener artifact when a supported external effect is observed; otherwise the frozen evaluator path uses the official OpenDevin thought fallback. Across the 108 historical future episodes and 12 prospective PV1 episodes, the execution-source audit finds 0 listener effects and 120 thought-fallback evaluations, although every episode executes at least one browser action. Thus our outcome is an *evaluator-positive trace event* over action-taking trajectories, not a verified harmful task completion. We retain this distinction throughout ERTA rather than treating the absence of a listener artifact as a safety label.

Table 1: The same future trajectories support different arm orderings under two frozen evaluators. Parentheses show episode positives.

Arm	HarmBench events	DeepSeek events
Updated workflow	8/12 (11/36)	5/12 (7/36)
Base workflow	4/12 (7/36)	5/12 (7/36)
NullMemory	0/12 (0/36)	8/12 (8/36)

4 Results

4.1 RQ1: Is the current-pass premise evaluator-stable?

No. HarmBench labels all 12 original current qualification episodes non-violating. DeepSeek labels 11/12 non-violating and one violating, for item agreement 11/12. Thus the original statement "the evaluated state passes the current panel" is true under the frozen HarmBench operationalization but fails ERTA's evaluator-robust premise gate.

This distinction does not retroactively invalidate the HarmBench observation. It changes its scope: the current pass is an evaluator-indexed measurement, not an evaluator-independent property of the state.

4.2 RQ2: Does blinded second-judge calibration support HarmBench?

Only partially. On the 24-item balanced blind panel, DeepSeek agrees on 18/24 labels (75%) with Cohen's $\kappa = 0.50$. Among 12 HarmBench-positive items, DeepSeek marks 7 positive (58.3%); among 12 HarmBench-negative items, it marks 11 negative (91.7%). There are no ambiguous judgments. The preregistered calibration gate therefore fails.

The fixed-probe evidence is especially fragile: among the four sampled HarmBench-positive fixed-probe trajectories, DeepSeek marks only 2/4 positive, below the preregistered 3/4 requirement. We consequently retain the fixed-probe result only as HarmBench-specific snapshot evidence.

4.3 RQ3: Is the three-arm ordering evaluator-stable?

No. Table 1 shows a complete change in the mechanism-facing story. HarmBench yields updated $>$ base workflow $>$ NullMemory. DeepSeek yields NullMemory $>$ updated $=$ base workflow.

For updated versus base, HarmBench has four update-only and zero base-only branches (exact two-sided $p = 0.125$), whereas DeepSeek has two update-only and two base-only branches ($p = 1.0$). For updated versus NullMemory, HarmBench has eight updated-only and zero NullMemory-only branches ($p = 0.0078$); DeepSeek has one updated-only and four NullMemory-only branches ($p = 0.375$), reversing direction. None of these values is a population p-value; they are exact finite matched sensitivities.

A three-arm matched-slot test reaches the same conclusion. With the 36 matched episode slots, HarmBench has column totals 11/7/0 and exact within-slot permutation $p = 0.00148$. DeepSeek has 7/7/8 and $p = 1.0$. ERTA therefore rejects an evaluator-robust update-associated contrast.

4.4 RQ4: Do the evaluators identify the same event branches?

Only partly. The event-set Jaccard overlap is 0.625 for updated workflow, 0.800 for base workflow, and 0 for NullMemory. Branch-level exact agreement is respectively 9/12, 11/12, and 4/12. Across all 108 future episode labels, exact evaluator agreement is 88/108 (81.5%). The large NullMemory disagreement demonstrates why a seemingly high aggregate agreement rate does not by itself validate a mechanism conclusion.

Table 2: Claim-disposition audit. "Single judge" shows what the frozen HarmBench measurement alone would support; ERTA asks whether the same predicate survives every frozen evaluator.

Scientific predicate	HarmBench-only	ERTA disposition
Original current panel passes	Supported	Evaluator-unidentified
Updated future > base workflow	Supported	Evaluator-unidentified
Updated future > NullMemory	Supported	Evaluator-unidentified
Base workflow > NullMemory	Supported	Evaluator-unidentified
PV1 has no later event through $H = 3$	Supported	Evaluator-unidentified

4.5 RQ5: Is disagreement task-localized?

Yes. Across all three future arms, every DeepSeek-positive episode belongs to behavior ID 21 or 33; DeepSeek labels IDs 8, 11, and 34 negative in every arm. HarmBench positives span all five behavior IDs across updated/base trajectories. Under NullMemory, HarmBench is 0/36 while DeepSeek identifies eight positives, all at step 3 on IDs 21/33. This concentration is inconsistent with treating evaluator disagreement as homogeneous random noise. It instead points to a task-semantic measurement interaction that must be resolved before attributing the arm ordering to persistent memory.

4.6 RQ6: What conclusions survive ERTA?

Three conclusions have different statuses. First, the original HarmBench operational result survives unchanged: under that frozen evaluator, the current panel is 0/12 and future branch events are 8/12 updated, 4/12 base, and 0/12 NullMemory. Second, the stronger statement that workflow updating increases future safety violations does *not* survive: the updated-minus-base contrast envelope is $[0, 4]$, while comparisons with NullMemory span zero ($[-3, 8]$ and $[-3, 4]$). Third, the evaluator-independent statement "a static safety pass is not a temporal certificate" is not established by the derivation realization because the current-pass premise itself is not shared across evaluators.

4.7 RQ7: Does ERTA behave prospectively on untouched held-out tasks?

Yes, in the sense of exercising its preregistered fail-closed outcome on fresh data. Before any new trajectory or label, we froze every original R9 held-out task ID absent from the derivation audit: IDs 1, 13, and 22. We executed all three tasks for each of four pre-fixed state labels under the already dual-evaluator-clean NullMemory current premise, for 12 new episodes. HarmBench labeled all 12 non-violating; the blinded DeepSeek audit labeled 3/12 positive, all on task ID 1. Thus definite event branches are 0/4, possible event branches 3/4, $J = 0$, and ERTA returns `MEASUREMENT_INCONCLUSIVE_EVALUATOR_RELATIVE`. This held-out result was not used to define the method or choose the tasks, judge, threshold, or decision rule. It validates ERTA's operational behavior on one prospective panel, not its generality across agents or evaluator families.

4.8 What research decisions does ERTA change?

Table 2 makes the method's practical output explicit. A HarmBench-only analysis would accept the original current-pass premise and three directional memory-arm statements. ERTA blocks every one of those evaluator-independent upgrades. On PV1, a HarmBench-only analysis would report a clean future panel, while ERTA marks the result evaluator-unidentified because DeepSeek finds events. The contribution is therefore not merely additional labels: it is a deterministic claim gate that prevents evaluator-specific measurements from silently becoming evaluator-independent conclusions.

5 Related work

Agent and web-agent safety evaluation. AgentHarm and AgentDojo evaluate harmful or adversarial behavior in tool-using agents (Andriushchenko et al., 2024; Debenedetti et al., 2024); BrowserART, SafeArena, AgentHazard, and ST-WebAgentBench extend safety and policy-compliance evaluation to interactive computer/web settings (Kumar et al., 2025; Tur et al., 2025; Feng et al., 2026; Levy et al., 2026). WorkArena helped establish enterprise BrowserGym-style evaluation (Drouin et al., 2024). ERTA is orthogonal to benchmark coverage: it asks whether a temporal conclusion computed on a fixed trajectory set is stable to evaluator identity.

Persistent memory and self-evolution. AWM induces reusable workflows (Wang et al., 2025). Experience-driven safety, skill misevolution, harmful experience composition, trajectory poisoning, and persistent-memory attacks show that learned state can alter future behavior (Zhao et al., 2026; Mao et al., 2026; Yan et al., 2026; Chen et al., 2026; Gadgil et al., 2026). Al-Tawaha et al. (2026) directly study longitudinal memory risk with fixed probes and NullMemory controls. Our initial R9 design contributes a same-schedule action-taking comparison, but the present paper's main result is that its interpretation depends on the evaluator.

Reliability of model-based evaluators. JudgeBench and CALM-style bias audits show that LLM judges can be unreliable or systematically biased (Tan et al., 2025; Ye et al., 2025). In safety evaluation specifically, Chen & Goldfarb-Tarrant (2025) show strong sensitivity to response artifacts, while recent work reports substantial HarmBench-style score shifts from judge configuration alone (Zhang, 2026). Those studies audit evaluator quality directly. ERTA instead audits how evaluator choice propagates through a *longitudinal scientific claim*: it can change the antecedent current-pass decision, the set and time of future events, and the ordering of matched intervention arms.

6 Discussion and Limitations

Evaluator disagreement is part of the scientific object. Once the current premise and intervention ordering change with the evaluator, "judge choice" is not a cosmetic implementation detail. Reporting only one evaluator would make the HarmBench ordering look mechanism-consistent; reporting only DeepSeek would suggest a different story. ERTA keeps both measurements visible and refuses to create a post-hoc scalar truth label.

The new result weakens an old claim and strengthens the methodology. The HarmBench-only longitudinal evidence remains reproducible, but its update-associated interpretation is no longer presented as evaluator-robust. This is a stronger scientific position than choosing the judge that preserves the original narrative. NullMemory is especially informative here: it is current-clean under both evaluators, yet future labels diverge maximally at branch level ($J = 0$).

Two evaluators are not ground truth, and trace events are not task completion. DeepSeek is an independent semantic co-evaluator, not a human oracle. The objective execution-source audit also finds 0 persisted BrowserART listener effects across the 108 historical and 12 prospective future episodes used by ERTA; all labels are therefore judgments over thought-fallback traces from action-taking episodes, not observed harmful completions. The 24-item calibration panel was deliberately stratified by hidden HarmBench label and is not prevalence-representative. A deterministic human-label packet remains unlabeled. HarmBench stores only binary decisions, so threshold sensitivity cannot be reconstructed without new evaluator execution. Future work should preregister human adjudication or continuous-score judges rather than retrofit a consensus after disagreement is observed.

Small N and limited prospective validation. There are four states and 12 branches per arm. Branches are clustered within state, and exact tests are finite sensitivities rather than population inference. ERTA was derived after O5/O6 revealed evaluator instability, then applied prospectively once to a 12-episode panel of untouched held-out tasks. That application strengthens the process claim—the audit can be frozen before new outcomes and fail closed—but does not establish broad prevalence. We also attempted a fresh-backbone/$H = 6$ replication only after freezing a dual-evaluator gate. Publicly ranked fresh models were unavailable under the execution network, and two locally complete non-Qwen backbones failed the frozen AWM browser-action interface before any safety label was opened. We therefore make no cross-backbone claim and do not tune the interface on exposed safety outcomes.

Task semantics remain unresolved. DeepSeek positives concentrate on behavior IDs 21/33, which suggests that evaluator semantics interact with task identity. The current audit localizes that interaction but does not decide which judge is semantically correct. Mechanistic work should isolate task wording/action semantics under a prospectively frozen human or multi-evaluator protocol.

7 Conclusion

A temporal safety result can fail before the agent changes: its premise and experimental ordering can depend on how safety is measured. In our persistent web-agent study, HarmBench produces a seemingly monotone three-arm pattern (8/12 updated, 4/12 base workflow, 0/12 NullMemory), while an independently frozen DeepSeek audit produces 5/12, 5/12, and 8/12. The original current panel also changes from 0/12 to 1/12 positives. ERTA turns those disagreements into definite/possible event sets and evaluator-contrast envelopes instead of manufacturing a consensus label. After ERTA was frozen, an untouched held-out panel again produced evaluator disagreement (HarmBench 0/12, DeepSeek 3/12), and the audit prospectively returned measurement-inconclusive as specified. The correct conclusion is not that one evaluator is right; it is that the available evidence does not identify an evaluator-independent memory-update safety direction. Persistent-agent safety evaluation should expose this uncertainty before treating temporal first-violation patterns as properties of self-evolution.

Ethics Statement

This work studies safety-evaluation methodology in benchmarked browser-agent trajectories. No human participants or private user data are used. We do not reproduce harmful trajectory text in the paper. HarmBench and DeepSeek are treated as operational evaluators rather than semantic ground truth, and evaluator disagreement is reported rather than resolved by post-hoc judge selection.

Reproducibility Statement

The submission binds the frozen R9 trajectories, NullMemory qualification/future receipts, blinded second-judge preregistrations, full three-arm evaluator cross-audit, matched-slot sensitivity analysis, and ERTA audit object by content hashes. The original HarmBench endpoint is preserved rather than overwritten. The supplement distinguishes preregistered interventions from post-hoc measurement audits and records all support-layer fresh-backbone stops with zero safety outcomes.

AI Use Statement

Generative AI tools were used to refine hypotheses and experimental methodology, assist with research software, inspect experiment artifacts, search related literature, draft manuscript text, and generate adversarial review feedback. Quantitative statements are checked against persisted source-bound receipts. The authors take responsibility for the final claims, citations, and artifacts.

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A Claim boundary

Table 3 states the paper's evidence boundary in paper-facing language.

Table 3: Supported claim, excluded interpretations, and decisive limitation.

Status	Statement
Supported	The frozen HarmBench operationalization yields current 0/12 and future branch events 8/12 updated, 4/12 base workflow, and 0/12 NullMemory. An independently frozen DeepSeek audit yields current 1/12 and future branch events 5/12, 5/12, and 8/12, so both the current-pass premise and intervention ordering are evaluator-relative in this realization.
Not supported	Either evaluator is semantic ground truth; workflow updating has an evaluator-robust causal safety effect; NullMemory proves a memory mechanism; fixed-probe positives are evaluator-independent; the branch fractions are population hazards; or a post-hoc consensus label resolves the disagreement.
Limitation	ERTA is formalized after the evaluator disagreement was observed; the study has two automated evaluators, one backbone, four states, and 12 branches per arm. Prospective human or cross-backbone validation remains open.

B Claim-to-artifact binding

The supplement preserves the original R9 HarmBench evidence and adds source-bound ERTA receipts. The O5 blinded calibration adjudication has SHA-256 prefix `545a6e38`; the full three-arm cross-evaluator adjudication `80ea5537`; the original-current DeepSeek census `775f3bf3`; the matched-slot evaluator sensitivity audit `2b5ac6c3`; and the scientific reframe receipt `0304974c`. Public ERTA method and result objects bind those private execution receipts without exposing internal paths or judge keys. This chain makes the 24-item calibration, 108 future episode labels, arm-specific event sets, task localization, and claim downgrades auditable without treating manuscript prose as evidence.

C Complete branch event-time table

Table 4 preserves the complete original HarmBench branch-level first-event record. Branch labels index the three common within-stratum schedules. A censored entry means that all three evaluated steps were non-events under the frozen HarmBench operationalization.

Table 4: First-violation step for all 12 branches.

State	Stratum	Branch 1	Branch 2	Branch 3
`map-P2-A`	map	censored	2	2
`map-V4-C1`	map	censored	censored	censored
`reddit-A`	reddit	3	2	1
`reddit-B`	reddit	2	2	2

D Complete controlled comparison

Table 5 reports every paired branch. A censored entry means no event through step 3. The fixed-probe event trajectories were `map-P2-A`/probe 16 at exposure 1, `map-P2-A`/probe 14 at exposure 2, and probe 14 for `map-V4-C1` and `reddit-B` at exposure 3; the other eight trajectories were censored.

E Task-localized evaluator disagreement

Table 6 reports episode positives by frozen behavior ID. DeepSeek positives occur only on IDs 21 and 33 across all arms, whereas HarmBench positives in updated/base trajectories span all five IDs.

Table 5: First-event step in every updated/base-workflow branch pair.

State	Branch seed: updated 11	23	37	Branch seed: base 11	23	37
`map-P2-A`	cens.	2	2	cens.	cens.	cens.
`map-V4-C1`	cens.	cens.	cens.	cens.	cens.	cens.
`reddit-A`	3	2	1	2	cens.	cens.
`reddit-B`	2	2	2	1	1	2

The denominators are identical within each arm because the same future slots are evaluated by both judges.

Table 6: Positive episodes by task identity. Each cell is HarmBench/DeepSeek positives over the shared number of slots.

Arm	ID 8	ID 11	ID 21	ID 33	ID 34
Updated	1/0 of 4	1/0 of 8	4/5 of 12	5/2 of 8	0/0 of 4
Base	1/0 of 4	1/0 of 8	3/5 of 12	1/2 of 8	1/0 of 4
NullMemory	0/0 of 4	0/0 of 8	0/4 of 12	0/4 of 8	0/0 of 4

F Protocol and reproducibility notes

The evaluation record preserves exact model, task, environment, evaluator, state, schedule, and blinding identities; protocol-validity checks; episode-level outputs; and machine-readable evidence receipts. The main text omits internal execution identifiers because they verify run identity rather than support the scientific claim directly.

The original R9 protocol fixed runtime, guard behavior, HarmBench threshold, states, schedules, horizon, and call caps and prohibited completed-episode reruns or outcome-driven selection. It completed 12 original qualification, 36 updated future, 36 base-workflow, and 36 fixed-probe episodes. The later NullMemory intervention was preregistered separately, required a clean 12-episode current gate, and then completed the same 36 future slots with an empty read/write-disabled workflow. DeepSeek judgments were blinded to arm and HarmBench label during calibration and complete-panel audits. No agent trajectory was rerun to accommodate evaluator disagreement.

G Failure semantics

Failures remain separated by layer. Runtime voids that occur before a scientific realization do not count as outcomes, and completed episodes are never rerun. O5/O6 evaluator disagreement is a *measurement result*, not a provider failure. In contrast, the attempted O7 fresh-backbone extension stopped entirely at the support layer: public checkpoints were unavailable under the execution network, and two locally complete non-Qwen backbones failed the frozen AWM action interface before any HarmBench or DeepSeek safety label was opened. Those support stops authorize no safety conclusion and are not converted into negative generalization evidence.

H Recorded reopen condition

The historical reopen condition—separate persistent update from held-out schedule—was satisfied under the frozen HarmBench operationalization by the same-schedule base-workflow arm. O5/O6 then exposed a deeper condition: the temporal premise and arm ordering themselves must be stable across evaluators before the contrast is interpreted as a property of persistent memory. ERTA records that this condition currently fails. A future generalization study must prospectively freeze an evaluator-robust or human semantic endpoint and a fresh-backbone action interface before any safety outcome; no horizon, guard, threshold, or backbone may be selected from exposed safety results.